THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 10

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** $(-7,\ 2)$	$B = (2x_M - x_A,\ 2y_M - y_A)$
2	**(d)** 13 units	$\sqrt{(3-(-2))^2 + (-9-(3))^2} = \sqrt{25+144} = 13$
3	**(b)** positive slope	values increase with x
4	**(b)** -5	Brahmagupta's rules for signs
5	**(b)** 2	$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$
6	**(a)** 25 cm	$r = \sqrt{24^2+7^2} = 25$
7	**(c)** 120 cm^2	area $= bh$
8	**(a)** 0.65	$1 - 0.35$
9	**(b)** 64	$ar^{n-1} = 1 \times (-2)^6 = 64$
10	**(c)** 9	it satisfies the 'if' part but not the 'then' part
11	**(a)** Both A and R are true and R is the correct explanation of A	satisfies 'if', fails 'then'
12	**(c)** 65	$\frac{40\times 60 + 10\times 85}{40+10} = 65$
13	**(c)** A is true but R is false	work is proportional to the value, not to the number of digits
14	**(d)** empty	we do not yet know any divisor
15	**(a)** centroid	Theorem 8
16	**(b)** if the opposite angles of a quadrilateral are equal, it is a parallelogram	Theorem 3
17	**(b)** (4, -1)	substitute the pair in both equations
18	**(c)** a straight line	plotted solutions lie on a line
19	**(c)** 308 cm^2	$2\pi r^2$
20	**(b)** 96 cm^2	side $= 4$ cm, TSA $= 6a^2 = 96$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$(-2,5)$: Quadrant II; $(6,-3)$: Quadrant IV; $(0,-4)$: on the y-axis; $(-1,-1)$: Quadrant III	½ × 4
22	Divide the unit interval $[0,1]$ into 4 equal parts and move 3 parts to the right of 0.	2
23	Chord at 3 cm: $2\sqrt{25-9} = 8$ cm; chord at 4 cm: $2\sqrt{25-16} = 6$ cm. The chord nearer the centre is longer.	1+1
24	Perimeter $= 2\pi r$, so the ratio of perimeters equals the ratio of radii: $5:4$.	2
25	$64 = 2^6$ has factors $1, 2, 4, 8, 16, 32, 64$: 7 factors (1). Odd number of factors, and 64 is a perfect square (8×8) (1).	1+1
26	$60 = 16\times 3 + 12$; $16 = 12\times 1 + 4$; $12 = 4\times 3 + 0$ (1). gcd $= 4$ (1).	1+1
27	True (1). Adjacent angles add up to 180° and opposite angles are equal, so each angle is 90° (1).	1+1
28	Volume $= 2\times 1\times 1.5 = 3$ m^3 (1) $= 3000$ litres (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Join diagonal AC to split ABCD into $\triangle ABC$ and $\triangle ACD$ (1). Each triangle has angle sum 180° (1). The angles of the two triangles together make up the four angles of ABCD: total $= 360^\circ$ (1).	1+1+1
30	Let y litres be added: $\frac{0.05\times 6 + 0.20y}{6+y} = 0.08$ (1). $0.3 + 0.2y = 0.48 + 0.08y$ (1), $0.12y = 0.18$, $y = 1.5$ litres (1).	1+1+1
31	$V = 77h$: $385,\ 693,\ 1001$ cm^3. Consecutive differences are equal (308) — a linear pattern.	2+1
32	$6x + 12 = 180 \Rightarrow x = 28$ (2). $\angle A = 108^\circ, \angle C = 72^\circ$ (1).	2+1
33	Let $P(0,y)$. $PA^2 = 36 + (y-5)^2$, $PB^2 = 16 + (y-3)^2$. $36 + y^2 - 10y + 25 = 16 + y^2 - 6y + 9 \Rightarrow 61 - 10y = 25 - 6y \Rightarrow y = 9$. $P = (0,9)$	1+1+1

34	(i) $64 + 225 = 289 = 17^2$ ✓ right-angled (1½). (ii) $100 + 576 = 676$ but $27^2 = 729$ × (1½).	1½+1½
35	Only 2 is both even and prime ⇒ $\frac{1}{6}$ (1). The numbers that are even or prime are 2, 3, 4, 5, 6, so only 1 is neither ⇒ $\frac{1}{6}$ (2).	1+2
	OR: Subjective: a person's own opinion from what they see ('sunny, so unlikely'), differing between people (1). Objective: based on evidence such as past rainfall data (relative frequency) or on equally likely outcomes (theoretical) (2).	1+2
36	New edge $= 1.5a$ (1). (i) TSA becomes $2.25\times$ the original: increase 125% (1). (ii) Volume becomes $3.375\times$ the original: increase 237.5% (1).	1+1+1
	OR: $V = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 10.5^3$ (1) $= 2425.5$ cm^3 (1) $= 2.4255$ litres (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Let medians BQ, CP meet at M; X, Y the midpoints of BM, CM (1). $PQ \parallel BC$, $PQ = \frac{1}{2}BC$ and $XY \parallel BC$, $XY = \frac{1}{2}BC$ (Midpoint Theorem in $\triangle ABC$ and $\triangle MBC$) (1). So $\triangle MPQ \cong \triangle MYX$ (ASA) ⇒ $MQ = MX = XB$ and $MP = MY = YC$ (1). Hence $CM : MP = 2 : 1 = BM : MQ$ (1). The same argument for BQ and AR gives the same point and ratio ⇒ concurrent (1).	1+1+1+1+1
	OR: P, Q, R, S midpoints of AB, BC, CD, DA: in $\triangle ABC$ and $\triangle ADC$, $PQ \parallel AC \parallel SR$; in $\triangle BCD$ and $\triangle BAD$, $QR \parallel BD \parallel PS$ ⇒ parallelogram (3). If $AC = BD = 16$ cm all sides equal 8 cm: a rhombus (1), perimeter 32 cm (1).	3+1+1
38	Let boat speed be x and stream speed y (km/h). Downstream speed $x + y = \frac{20}{2} = 10$ (1). Upstream speed $x - y = \frac{12}{2} = 6$ (1). Adding: $2x = 16$, $x = 8$ (1). Then $y = 2$ (1). Boat 8 km/h, stream 2 km/h; check $8 + 2 = 10$, $8 - 2 = 6$ (1).	1+1+1+1+1
	OR: $l + w = 22$ (1). New sides equal: $l - 2 = w + 3$, so $l - w = 5$ (1). Adding: $2l = 27$, $l = 13.5$ cm (1). $w = 8.5$ cm (1). Area $= 13.5 \times 8.5 = 114.75$ cm^2 (1).	1+1+1+1+1
39	(a) AP, $d = 4$, $t_n = 4n + 3$ (1). (b) GP, $r = 3$, $t_n = 2 \cdot 3^{n-1}$ (1). (c) Neither (differences 3,5,7; ratios 4, 2.25) but $t_n = n^2$ (1). (d) GP, $r = \frac{1}{2}$, $t_n = 100\left(\frac{1}{2}\right)^{n-1}$ (2).	1+1+1+2
	OR: (i) 100, 150, 200 (1). (ii) $t_n = 100 + 50(n-1) = 50n + 50$ (1). (iii) $t_{12} = 650$; total $= \frac{12}{2}(100 + 650) = ₹4,500$ (2). (iv) $50n + 50 = 1000 \Rightarrow n = 19$ (1).	1+1+2+1
40	(i) $x^2 + 8x + 15 = (x + 3)(x + 5)$. (ii) $x(x - 4) = 96 \Rightarrow (x - 12)(x + 8) = 0$, so $x = 12$ m, breadth 8 m. (iii) $2(12 + 8) = 40$ m.	1+3+1
	OR: (i) $100 = 38 + 2(xy + yz + zx) \Rightarrow 31$. (ii) $x^3 + y^3 + z^3 - 3xyz = (10)(38 - 31) = 70 \Rightarrow x^3 + y^3 + z^3 = 70 + 75 = 145$.	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.